

Muscle Contraction as Ion-Geometry Coupling: A Field-Geometry Framework

A Theoretical Proposal for the Geometric Basis of Cross-Bridge Mechanics, Fatigue, and Restoration

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Abstract

Conventional biochemical accounts of muscle contraction treat the cross-bridge cycle as a chemical motor: ATP hydrolysis provides energy, myosin binds actin, a conformational change produces force, and the cycle repeats. This framing is not wrong — but it is incomplete. Chemistry sets the energy budget. Geometry determines how that budget is spent. This paper argues that the cross-bridge cycle is more precisely understood as a sequence of geometric state changes in the actin-myosin lattice, triggered by ion-mediated tension injection, propagated as a geometric wave, reset by ATP-driven geometry restoration, and disrupted by geometry drift. The core claim is this: contraction is a geometric event that chemistry powers. Actin and myosin are spatially organized field-geometry components whose productive interaction depends on alignment within nanometer-scale geometric tolerance windows. The action potential delivers not merely a trigger but an ion wave that physically shifts the electrostatic field geometry of the thin filament, displacing tropomyosin and opening contraction windows. The transmembrane EMF is a structural component of muscle geometry, not merely a signaling threshold. ATP's primary function in the cross-bridge cycle is geometric reset, not energy donation for the power stroke. Fatigue represents geometry drift — progressive ionic and structural departure from the range that supports productive cross-bridge formation — rather than simple energy depletion. This framework unifies contraction, fatigue, spasm, and therapeutic restoration under a single geometric language and identifies testable predictions that distinguish it from purely chemical models.

1. Contraction Is Geometry, Not Chemistry

The standard account assigns causality to ATP hydrolysis and calls the result motion. But motion is geometry changing over time. That is not a metaphor — it is a definition. What actually moves during muscle contraction? Myosin heads rotate through a defined angular arc of approximately 70 degrees [1]. Actin filaments translate a defined displacement of roughly 5–10 nm per cross-bridge stroke [2, 3]. These are geometric facts with spatial coordinates and angular measurements attached. Chemistry provides the energy. Geometry dictates what that energy does.

The problem with purely chemical descriptions of contraction is this: they can predict energy output but they cannot predict spatial specificity or directionality without silently invoking structural geometry. When a biochemist says "myosin undergoes a conformational change," they are already describing geometry. When they say the power stroke "pulls" the thin filament, they are specifying a direction — a geometric vector. The chemical language borrows geometry constantly, then declines to name it.

This paper names it. The central proposal is that muscle contraction is best defined as a sequence of geometric state changes in the actin-myosin lattice, triggered and sustained by the ion field environment. The ***geometric state*** of a contracting fiber is the instantaneous configuration of the contractile lattice: the relative positions of thick and thin filaments, the angular disposition of myosin heads, the position of tropomyosin along the actin helix, and the electrostatic field geometry imposed by the local ion environment. At any moment, the geometric state either permits productive cross-bridge formation or it does not.

Chemistry — ATP hydrolysis, Ca^{2+} binding, Pi release — drives transitions between geometric states. But the transitions themselves, and the spatial outcomes they produce, are fully described only in geometric terms. Displacement, angle, distance, overlap, and periodicity are the primary language of contraction. This paper uses that language consistently.

Core Claim — Section 1

Contraction is a geometric event powered by chemistry. The causal hierarchy runs: ion field → geometric state change → mechanical output. ATP hydrolysis is not the cause of motion — it is the energy source for a sequence of geometric transitions whose directionality and specificity are determined by lattice geometry.

2. Actin and Myosin as Field-Geometry Components

Actin and myosin are not merely proteins that happen to be arranged in a regular pattern. They are spatially organized field-geometry components whose structure encodes the rules of their interaction. The geometry is not incidental to function. It *is* the functional unit.

2.1 The Actin Helix

Filamentous actin (F-actin) is a double-stranded helix with a crossover repeat of approximately 36 nm and a subunit repeat of approximately 5.5 nm [4]. Each actin monomer within the filament presents a defined binding face with a specific charge distribution — the myosin-binding cleft on actin carries a characteristic negative charge density that is geometrically matched to positively charged regions on the myosin head [5]. This is not incidental chemistry. The spatial periodicity of binding sites along the actin filament — one site every ~5.5 nm — defines the geometry of every possible cross-bridge interaction.

2.2 The Myosin Thick Filament

The myosin thick filament is equally precise. Myosin heads project outward from the thick filament backbone at regular angular intervals — approximately every 14.3 nm along the filament axis, arranged in a three-stranded helix with a repeat of 43 nm [6]. The heads project at defined angles relative to the filament axis. Their spatial relationship to surrounding actin filaments is determined by the lattice geometry of the sarcomere — not by diffusion or random encounter. In a resting sarcomere, the inter-filament spacing

(center-to-center distance between thick and thin filaments) is approximately 40–45 nm, and this spacing changes with sarcomere length and osmotic conditions [7].

2.3 Geometric Matching

Cross-bridge formation requires geometric matching. A myosin head binds to an actin site not because it "reaches" actin in some general sense, but because the geometric vectors of the head and the binding site align within a spatial tolerance that permits stereospecific docking [8]. The periodicity mismatch between actin (5.5 nm subunit repeat) and myosin (14.3 nm axial repeat) means that at any given moment, only a subset of myosin heads are geometrically aligned with actin binding sites. The rest are out of register. This periodicity mismatch is not a flaw in the system — it is what allows distributed, asynchronous cross-bridge cycling that produces smooth rather than stuttering force [6].

2.4 Tropomyosin and Troponin as Geometry Gatekeepers

Tropomyosin runs along the actin filament, occupying a groove on the helix and physically occluding myosin-binding sites when muscle is at rest [9]. Troponin — specifically the troponin-I subunit — stabilizes this blocking position. Troponin-C, when it binds Ca^{2+} , undergoes a conformational change that weakens the troponin-I interaction with actin, allowing tropomyosin to shift azimuthally by approximately 1.5 nm around the filament axis [10]. This shift is a geometric displacement. It exposes myosin-binding sites. It is not a chemical reaction in the energetic sense — it is a spatial relocation driven by the electrostatic consequences of Ca^{2+} binding.

Tropomyosin and troponin are geometry gatekeepers: they regulate access to actin-binding sites by controlling a geometric switch. The switch has two positions — blocked and open — and the transition between them is driven by the ion field, not by ATP.

2.5 The Sarcomere Lattice as a Three-Dimensional Geometric Field

The sarcomere is not a linear arrangement. It is a three-dimensional crystalline lattice in which each thick filament is surrounded by six thin filaments in a hexagonal array, and each thin filament is shared among three thick filaments [11]. This geometry means that

every myosin head operates within a defined spatial field — its reach, its angle, and its competitive interactions with neighboring heads are all geometrically constrained by the lattice. Disruptions to lattice geometry — through changes in osmotic pressure, sarcomere length, or filament spacing — alter the probability of productive cross-bridge formation across the entire lattice simultaneously.

3. Ion Influx as Tension Injection

The action potential does not simply "trigger" contraction. It delivers an ion wave that injects tension into the geometric lattice. The distinction matters. A trigger is binary — on or off. Tension injection is quantitative, spatial, and time-dependent. The magnitude, rate, and spatial distribution of the ion wave determine the geometry of the contraction event that follows.

3.1 The Electrochemical Gradient as Pre-Loaded Tension

Before a single action potential fires, the muscle fiber is pre-loaded. The resting membrane potential of skeletal muscle (~ -85 mV) represents a standing electrochemical disequilibrium — a store of ionic tension maintained by the Na^+/K^+ -ATPase and the selective permeability of the sarcolemma [12]. Na^+ is concentrated extracellularly. K^+ is concentrated intracellularly. Ca^{2+} is concentrated in the sarcoplasmic reticulum (SR) at concentrations three to four orders of magnitude higher than the resting cytosolic level (~ 100 nM at rest vs. ~ 1 mM in the SR lumen) [13]. These gradients are geometric potential energy — tension stored in ionic distributions across defined spatial boundaries.

3.2 Calcium as the Tension-Injecting Ion

Ca^{2+} is the primary tension-injecting ion. Its charge density (+2), its high binding affinity for troponin C ($K_d \sim 0.1$ – 1 μM in the physiological range), and the geometry of the troponin C binding cleft make it uniquely suited for this role [14]. When Ca^{2+} floods the cytosol from the SR — reaching peak concentrations of 10 – 100 μM during a twitch — it binds the EF-hand domains of troponin C and changes the electrostatic field geometry around the thin filament. The result is tropomyosin displacement (~ 1.5 nm), binding site exposure, and

cross-bridge formation. Ca^{2+} does not provide energy for this process. It provides the field geometry change that shifts the tropomyosin switch.

This is tension injection: the ion field shift translates into a mechanical bias across the thin filament lattice. The thin filament goes from a state of blocked geometry to a state of open geometry — not through bond breakage or energy expenditure, but through the spatial consequences of ion binding.

3.3 Spatial Spread via the T-Tubule System

The T-tubule network ensures that this tension injection is not confined to the fiber surface. T-tubules are sarcolemmal invaginations that penetrate the fiber interior, running transversely at each A-I junction (in mammalian skeletal muscle) and reaching within approximately $0.2\text{ }\mu\text{m}$ of every myofibril [15]. The action potential propagates down the T-tubule lumen at the same velocity as it travels along the surface ($\sim 2\text{--}5\text{ m/s}$), ensuring that Ca^{2+} release from the SR occurs nearly synchronously across the full fiber cross-section. The T-tubule geometry is a three-dimensional waveguide for tension injection — it converts a surface electrical event into a volumetric ion-field event that simultaneously opens contraction windows across thousands of sarcomeres arranged in parallel.

4. Contraction Windows

Contraction does not occur uniformly across a muscle fiber or a sarcomere. It occurs in windows — spatiotemporal zones where geometric alignment between actin and myosin reaches a threshold sufficient for productive cross-bridge formation. Understanding contraction means understanding where and when these windows open, how wide they are, and what closes them.

4.1 Defining the Contraction Window

A **contraction window** is the spatiotemporal interval during which Ca^{2+} concentration, filament geometry, and ion-field tension are simultaneously within binding-permissive ranges. All three conditions must be met. High Ca^{2+} without correct filament geometry produces no useful cross-bridge formation — this occurs, for example, when sarcomeres

are at extreme lengths and geometric overlap is lost. Correct filament geometry without sufficient Ca^{2+} leaves the troponin switch in its blocking configuration. Ion-field tension without geometric alignment produces futile attachment attempts. The contraction window is the intersection of these conditions in space and time.

4.2 The Overlap Zone as Geometric Boundary

The physical boundary of the contraction window is the overlap zone — the region within each half-sarcomere where thick and thin filaments actually co-occupy space and myosin heads can reach actin binding sites [16]. At optimal sarcomere length ($\sim 2.0\text{--}2.2\ \mu\text{m}$ in human skeletal muscle), overlap is maximal. At lengths shorter than $\sim 1.6\ \mu\text{m}$, thin filaments from opposite halves of the sarcomere interfere with each other, disrupting the geometric lattice. At lengths longer than $\sim 3.6\ \mu\text{m}$, thick and thin filaments no longer overlap, and no cross-bridges can form regardless of Ca^{2+} concentration [16]. The window has hard geometric boundaries.

4.3 The Length-Tension Relationship as Window Geometry

The length-tension relationship — one of the most replicated findings in muscle physiology — is the macroscopic expression of window geometry [16]. As sarcomere length decreases from optimal toward the ascending limb, window geometry deteriorates through thin-filament double-overlap and internal resistance from the myosin backbone encountering Z-discs. As sarcomere length increases beyond optimal toward the descending limb, window geometry deteriorates through loss of filament overlap. Force falls in both directions because windows shrink or close. The curve is not a property of the chemistry — it is a property of the geometry.

4.4 Tetanic Contraction as Window Stacking

A single twitch produces a brief, submaximal contraction because the contraction window opens and closes once — Ca^{2+} is released, diffuses, activates, and is then re-sequestered by the SR Ca^{2+} -ATPase (SERCA) within $\sim 50\text{--}100\ \text{ms}$ [13]. Tetanic contraction occurs when successive action potentials arrive before geometric reset is complete. Each new action potential injects additional tension into the lattice before the previous Ca^{2+} transient has

decayed. Windows stack. The cumulative effect is sustained, maximal Ca^{2+} saturation of troponin C, maximum tropomyosin displacement, and the highest possible cross-bridge density. Tetanus is not a different process — it is the spatial and temporal superposition of overlapping contraction windows.

5. Cross-Bridge Geometry Matching

A cross-bridge forms not because myosin "reaches" actin in some diffuse sense, but because the geometric vectors of the myosin head and the actin binding site align within spatial tolerance. This is stereospecific docking at nanometer resolution. It is geometry, not chemistry, that decides whether a given encounter between head and site produces a productive bond.

5.1 Stereospecificity at Nanometer Resolution

The crystal structure of the myosin S_1 fragment complexed with actin — resolved by Rayment and colleagues in 1993 — revealed that the actin-myosin interface involves approximately 7–8 actin residues making contact with a defined face of the myosin motor domain [8]. The interface is not just complementary in charge — it is complementary in three-dimensional shape. Rotational misalignment by even a few degrees substantially reduces the probability of stable bond formation. The binding is lock-and-key at nanometer resolution, with the lock being the actin binding groove and the key being the myosin lower 50 kDa domain surface.

5.2 The Lever Arm as Geometric Amplifier

The myosin S_1 region is structured as a motor domain connected to a lever arm — a long alpha-helical region stabilized by essential and regulatory light chains [17]. Small conformational changes at the nucleotide-binding pocket of the motor domain are amplified through the lever arm into large angular displacements at the lever arm tip. The power stroke involves a rotation of the converter domain by approximately 65–70 degrees, which — leveraged through the ~10 nm arm — produces a working stroke of approximately 5–10 nm at the actin attachment point [2, 3]. The lever arm is a geometric amplifier: it

converts molecular-scale bond rearrangements into filament-level displacements. Without this geometry, the chemical energy released by ATP hydrolysis would produce force without useful motion.

5.3 The Geometric Tolerance Window

Cross-bridge formation is probabilistic within a *geometric tolerance window*. If the myosin head is within spatial tolerance of an actin binding site — meaning the head's binding face is correctly oriented and the inter-filament distance is within reach — attachment probability is high. If the head is out of register along the helical pitch of actin, attachment probability drops sharply. If the inter-filament spacing is too large (as occurs at low osmotic pressure or extreme sarcomere lengths), heads cannot reach sites even when rotationally aligned. This tolerance window has been estimated from structural and mechanical studies to be on the order of ± 5 – 6 nm axially and ± 15 degrees rotationally [6, 18].

5.4 Distributed Geometry Matching Across the Thick Filament

A single thick filament carries approximately 300 myosin heads distributed across its length. Because of the periodicity mismatch between actin and myosin repeat distances, these heads are never all in register simultaneously — they are spread across many different phases of geometric alignment with the surrounding actin filaments [6]. This distribution is not a liability. It is the mechanism by which smooth, continuous force is generated. At any given moment, some heads are in the geometric tolerance window and forming or completing strokes, while others are out of register and idle. The thick filament is a distributed geometry-matching system, not a synchronized cannon.

5.5 Geometric Friction During Shortening

During rapid shortening, cross-bridges attached at non-optimal geometry impose a resistive force — geometric friction. A head that attached when the filament geometry was favorable becomes geometrically unfavorable as the filament slides past. The head is now being dragged through an off-axis orientation, and it resists the displacement vector. This is the mechanical basis of the shortening-velocity dependence of force (the Hill force-

velocity relationship) [19]: at high shortening velocities, the detachment rate cannot keep up with the rate at which attached heads are dragged into non-productive geometry. The muscle is fighting its own attached cross-bridges.

6. EMF-Geometry Coupling

The resting membrane potential of skeletal muscle (~ -85 mV) is not merely a threshold that the action potential must overcome. It is a structural component of muscle geometry. The transmembrane electric field imposed by this potential — approximately 10 million volts per meter across the ~ 10 nm membrane thickness — physically orients charged molecular structures within and adjacent to the membrane. Changing this field changes the geometry of those structures. This is EMF-geometry coupling.

6.1 The DHPR as EMF-Geometry Transducer

The dihydropyridine receptor (DHPR, or $\text{Ca}_v1.1$) is a voltage-sensing L-type calcium channel embedded in the T-tubule membrane. Its S_4 transmembrane segments contain positively charged arginine and lysine residues that respond to membrane potential change with a physical translocation — the "gating charge" movement [20]. In skeletal muscle, this movement involves the displacement of approximately 12–14 elementary charges across the electric field. This is not a chemical reaction. It is a direct geometric response to a change in the local electric field — a rotation and translation of charged segments driven by electrostatic force.

6.2 DHPR-RyR1: Direct Geometry Transmission

In skeletal muscle, the DHPR does not release Ca^{2+} by opening its own channel pore to any significant degree. Instead, the conformational change in the DHPR voltage-sensing domain is transmitted mechanically and directly to the ryanodine receptor type 1 (RyR1) on the adjacent SR membrane [21]. The two proteins are physically coupled — DHPR tetrads sit in direct apposition to every other RyR1 in the SR membrane — and the geometric change in the DHPR physically opens the RyR1 gate. This is geometry transmission: a conformational change in one protein, driven by an electric field, mechanically opens a

channel in another protein across the junctional gap. No Ca^{2+} -induced Ca^{2+} release (CICR) is required in skeletal muscle. The geometry of the DHPR-RyR₁ complex is the transduction mechanism.

6.3 Membrane Potential as a Standing Geometric Bias

At rest, the negative membrane potential holds the DHPR S₄ segments in their inward, "down" position — a configuration that keeps RyR₁ closed and maintains the geometric block on Ca^{2+} release [20]. Depolarization during the action potential moves the S₄ segments outward, removes the geometric block, and allows RyR₁ to open. The key point is this: depolarization removes suppression rather than actively driving contraction. The geometry is poised to produce contraction; rest is an actively maintained suppressed state. The electric field is not the initiator of contraction — it is the guard that prevents contraction from happening continuously.

6.4 Ionic Composition and EMF Geometry

Changes in extracellular ionic composition directly alter the EMF geometry and thus the contraction threshold. Hypokalemia (low extracellular K^+) hyperpolarizes the membrane, making action potential initiation harder and reducing contraction drive — the geometric guard is more firmly engaged [22]. Hyperkalemia (high extracellular K^+) depolarizes the resting membrane potential, which initially increases excitability but eventually inactivates Na^+ channels and impairs action potential propagation — the EMF geometry is disrupted in a way that collapses the action potential. Acidosis (H^+ accumulation) directly reduces Ca^{2+} sensitivity of troponin C, narrowing the contraction window without changing the EMF geometry per se [23]. Each of these ionic perturbations produces a distinct, geometry-consistent pattern of contraction impairment.

7. ATP as Geometry Reset

ATP's primary role in the cross-bridge cycle is not energy donation for the power stroke. It is geometry reset. The power stroke happens during P_i release — the transition from weak to strong actin binding, coupled to lever arm swing [17]. ATP binding to myosin *after* the

stroke breaks the geometric lock between myosin and actin and enables the head to return to pre-stroke geometry. Remove ATP, and the geometry freezes. That frozen state is rigor.

7.1 The Geometry Reset Sequence

The cross-bridge cycle, understood geometrically, proceeds in five steps:

1. **ATP binds myosin.** Myosin-actin binding affinity drops by approximately three orders of magnitude. The head detaches from actin — the geometric lock is released.
2. **ATP hydrolysis cocks the lever arm.** Hydrolysis of ATP to ADP·Pi at the nucleotide pocket drives the lever arm back to its pre-stroke ("cocked") position — an angular displacement storing approximately 30 kJ/mol as geometric potential energy in the strained lever arm conformation [17].
3. **The cocked head re-engages actin weakly.** This is a low-affinity, geometrically exploratory interaction. The head is in the correct geometry to dock with an actin site if it encounters one within tolerance.
4. **Pi release triggers the power stroke.** Pi dissociation from the active site is tightly coupled to the isomerization from weak to strong actin binding and the lever arm swing — the cocked geometry relaxes, converting stored geometric potential into filament displacement [17].
5. **ADP release locks post-stroke geometry.** With ADP gone, the head occupies a high-affinity, post-stroke geometric state. It remains bound to actin until ATP arrives to begin the cycle again.

7.2 Rigor Mortis as Geometry Lock

Rigor mortis is not a chemical curiosity — it is a direct demonstration that ATP's role is geometric reset, not power stroke energy. After death, metabolic ATP production ceases. Existing ATP is consumed as remaining cross-bridge cycles run to completion. With no ATP to bind post-stroke myosin heads, those heads cannot detach. They lock in post-stroke

geometry — strongly bound, lever arm swung, non-cycling [24]. Every sarcomere becomes mechanically rigid because no head can reset. The resulting stiffness — rigor — confirms that ATP's function in breaking the actin-myosin bond is as essential as its role in energizing the stroke.

7.3 Geometric Potential Stored by Hydrolysis

The ~30 kJ/mol released by ATP hydrolysis is not primarily dissipated as heat during the reset — it is stored as geometric potential in the strained, cocked lever arm conformation [17]. The lever arm in the pre-stroke position is mechanically strained relative to the post-stroke position, and this strain is the physical embodiment of the energy that will be released as the power stroke. ATP hydrolysis is a geometry-charging event. The power stroke is the geometry discharge.

7.4 Reset Rate as Contraction Pacemaker

Maximum shortening velocity is limited not by force generation capacity but by how fast geometry can be reset across the cross-bridge population. The rate-limiting step for shortening velocity is ADP release (which must precede ATP binding) and the kinetics of the subsequent detachment and re-cocking sequence [19]. At maximum shortening velocity, the muscle reaches the condition where heads are detaching and re-cocking as fast as the cycle allows — the shortening velocity at which force approaches zero ($V_{\max}$) is a direct measurement of the maximum geometry reset rate for a given myosin isoform.

8. Contraction Propagation

A muscle does not contract all at once. Contraction propagates — it spreads as a geometric activation wave driven by the action potential traveling along the sarcolemma and into the T-tubule network. The spatial and temporal properties of this propagation determine when and where contraction windows open across the fiber.

8.1 Action Potential Propagation as Activation Front

The action potential propagates along the muscle fiber surface at approximately 2–5 m/s [12, 25]. This conduction velocity is the propagation rate of the geometric activation front

— the leading edge of depolarization that sequentially opens DHPR voltage sensors, triggers RyR1, and initiates Ca^{2+} release along the fiber length. A 10-cm muscle fiber is traversed in approximately 20–50 ms. This means that sarcomeres at opposite ends of the fiber do not activate simultaneously — they activate with a small but real temporal offset determined by conduction velocity and fiber length.

8.2 T-Tubule Geometry as Three-Dimensional Waveguide

The T-tubule network solves a critical geometry problem: how does the surface action potential reach the interior myofibrils without a prohibitive propagation delay? The answer is the T-tubule architecture — a three-dimensional network of sarcolemmal invaginations that distributes the depolarization wavefront throughout the fiber volume within milliseconds [15]. The T-tubule system acts as a waveguide, ensuring that DHPR voltage sensors in the fiber interior receive the same temporal signal as those at the surface. Without this geometry, central myofibrils would activate significantly later than peripheral ones, producing non-uniform contraction and reduced force output.

8.3 Motor Units as Propagation Domains

A motor unit — one motor neuron and all the muscle fibers it innervates — is the fundamental propagation domain of contraction. All fibers within a motor unit receive the same neural signal simultaneously and activate within milliseconds of each other. Different motor units, however, may activate asynchronously — a property exploited by the nervous system for graded force production through asynchronous motor unit recruitment [26]. The temporal staggering of propagation domains across a muscle means that the muscle's total force output is the spatial and temporal sum of many partially overlapping geometric activation events.

8.4 Summation as Geometric Superposition

Graded force production through recruitment summation is geometric superposition: more motor units activated means more propagation fronts active simultaneously, more contraction windows open in parallel, and greater total cross-bridge density. Rate coding — increasing the firing frequency of already-active motor units — produces temporal

superposition: successive contraction events overlap before geometric reset is complete, stacking windows and increasing force toward the tetanic maximum.

8.5 Eccentric Contraction as Reversed Propagation Geometry

Eccentric contraction — muscle lengthening under load — is the geometric inverse of concentric contraction. The activation front propagates normally, contraction windows open, and cross-bridges form. But external force pulls the sarcomere apart while the geometric activation attempts to shorten it. Cross-bridges that attached at favorable geometry are stretched into progressively less favorable geometry as the sarcomere lengthens. They resist the displacement vector — this is the source of the unusually high force produced during eccentric contraction (which can exceed isometric force by 50–100%) [27]. The geometry of the attached cross-bridges is being strained against the shortening vector, converting the geometric strain into a resistive force that opposes lengthening.

9. Muscle Failure as Geometry Drift

Fatigue and failure are not primarily energy depletion events. The characterization of fatigue as "running out of ATP" is not supported by the evidence — intracellular ATP levels rarely fall below 70% of resting values even at the point of task failure [28]. Fatigue is geometry drift: progressive departure of the ion field environment and filament geometry from the ranges that support productive cross-bridge formation. The drift is multi-site, cumulative, and self-compounding.

9.1 Intracellular Acidosis

H⁺ accumulation — a product of metabolic activity during intense exercise — reduces Ca²⁺ sensitivity of troponin C [23, 28]. The geometry switch that moves tropomyosin requires higher Ca²⁺ concentration to produce the same azimuthal displacement. The contraction window narrows: more Ca²⁺ is needed to open it, less Ca²⁺ is available to sustain it (because SERCA is also H⁺-sensitive), and the window is open for a shorter fraction of each twitch.

Force falls not because sarcomeres are damaged, but because geometry-permissive conditions are harder to achieve.

9.2 Inorganic Phosphate Accumulation

Pi accumulates during rapid ATP hydrolysis when rephosphorylation cannot keep pace with consumption. Elevated intracellular Pi rebinds to myosin heads that have completed the Pi-release step, driving them back toward pre-stroke geometry before the stroke is mechanically useful [28, 29]. The fraction of heads in productive geometric states drops. Force per cross-bridge decreases. Pi is a geometry disruptor — it interferes with the geometry reset sequence by reversing the most mechanically productive step.

9.3 K⁺ Accumulation in T-Tubule Lumen

Each action potential drives K⁺ out of the muscle fiber through voltage-gated K⁺ channels. In the confined T-tubule lumen, this K⁺ accumulates faster than it can diffuse away [22, 30]. The local K⁺ concentration rises, the transmembrane K⁺ gradient collapses, and the resting membrane potential depolarizes toward the K⁺ equilibrium potential. The EMF geometry shifts. Na⁺ channels inactivate at the depolarized membrane potential. Action potential amplitude decreases. Eventually, action potentials fail to propagate into the T-tubule lumen at all — and the Ca²⁺ release that depends on T-tubular DHPR activation does not occur. Contraction windows stop opening not because the sarcomere is damaged, but because the geometry of the EMF field that drives Ca²⁺ release has collapsed.

9.4 Sarcomere Length Non-Uniformity

Under prolonged or eccentric loading, sarcomeres within a single fiber drift to different lengths — a phenomenon documented extensively in single-fiber and whole-muscle studies [31]. Sarcomeres on the descending limb of the length-tension curve are geometrically unstable: a small additional stretch reduces force, which allows further stretch, which further reduces force. These sarcomeres elongate while adjacent sarcomeres compensate by shortening. The result is a population of sarcomeres spread across a wide length distribution, with many operating at non-optimal geometry. Some sarcomeres work against others. Total fiber force collapses not because all sarcomeres are failing, but because

the geometric distribution across the sarcomere population has drifted away from the coherent optimal state.

9.5 Geometry Drift: The Unified Concept

Geometry drift is the progressive departure of the ionic field environment, filament geometry, and EMF structure from the ranges that support productive cross-bridge formation. Each of the mechanisms above — acidosis, Pi accumulation, K^+ accumulation, sarcomere length non-uniformity — represents a different dimension of geometry drift. They compound each other. Acidosis impairs Ca^{2+} sensitivity while Pi impairs stroke efficiency while K^+ accumulation impairs activation while sarcomere non-uniformity impairs the overlap geometry. Small departures in each dimension multiply across millions of sarcomeres to produce macroscopic weakness without cellular death. The cell is intact. The geometry is not.

10. Restoring Geometry to Fix Contraction

If contraction failure is geometry drift, restoration requires geometry correction. This reframes the logic of recovery, rehabilitation, and therapeutic intervention. The question is not "how do we replenish energy?" — that is a secondary concern. The question is "how do we restore the ion field geometry, filament alignment, and EMF structure that support productive cross-bridge formation?"

10.1 Active Recovery as EMF Geometry Restoration

Low-intensity activity during recovery — the physiological basis of "active cooldown" — flushes accumulated K^+ from T-tubule lumens back into the interstitium through diffusion and lymphatic clearance, driven by the concentration gradient [22, 30]. Continued moderate Na^+/K^+ -ATPase activity restores the transmembrane K^+ gradient. The resting membrane potential returns toward -85 mV. The EMF geometry of the T-tubule system is restored, and action potential propagation into the fiber interior becomes reliable again. Active recovery restores geometry. Passive rest restores it more slowly.

10.2 Stretching as Sarcomere Length Redistribution

Passive stretching redistributes sarcomere lengths within a fiber toward greater uniformity by applying a macroscopic length change that is shared across the sarcomere population [31]. Sarcomeres that drifted to extreme lengths during fatiguing eccentric activity are brought back toward the plateau of the length-tension curve. The fraction of sarcomeres operating in geometry-permissive length ranges increases. This is a direct geometric intervention — it acts on the spatial distribution of sarcomere lengths, not on the chemistry of the cross-bridge cycle itself.

10.3 Electrolyte Repletion as Field-Geometry Restoration

Restoring Na^+ , K^+ , Ca^{2+} , and Mg^{2+} to normal serum concentrations restores the ionic gradients that constitute the transmembrane tension store [12, 13]. K^+ repletion restores the resting membrane potential. Ca^{2+} repletion ensures SR loading capacity is maintained for the next contraction cycle. Mg^{2+} repletion matters because Mg^{2+} is required for MgATP — the actual substrate for myosin ATPase — and for modulating RyR1 gating [13]. Electrolyte repletion is field-geometry restoration: it rebuilds the ionic foundation that the contraction-window-opening process depends on.

10.4 Neuromuscular Electrical Stimulation as Geometry Bypass

Neuromuscular electrical stimulation (NMES) bypasses the upstream geometry — the motor neuron, the neuromuscular junction, the sarcolemmal action potential propagation — and directly depolarizes the muscle fiber membrane. It opens contraction windows by imposing depolarization directly, circumventing the K^+ -accumulation-driven EMF geometry failure that characterizes fatigue [32]. This is why NMES can elicit force in muscles that are neurologically inexcitable due to fatigue or pathology. The sarcomere geometry is intact; only the upstream electrical geometry has failed. NMES restores the driving signal without restoring the upstream geometry.

10.5 Muscle Spasm as Geometry Lock

Spasm is the converse of fatigue: instead of contraction windows failing to open, they fail to close. Sustained high intracellular Ca^{2+} — whether from impaired SERCA function,

membrane damage, or excessive neural drive — maintains tropomyosin in the open position and keeps cross-bridges continuously cycling [13]. The geometric lock of the post-stroke configuration is broken by ATP, but new cross-bridges immediately re-engage in the persistently open contraction window. The muscle cannot reset. Treatments that reduce Ca^{2+} availability (muscle relaxants that increase GABA-ergic inhibition upstream, or direct SERCA enhancement) or restore K^+ gradients (reducing excitability and action potential frequency) break the geometric lock by restoring the geometry of the relaxed state.

10.6 Chronic Dysfunction as a Geometry Problem

Chronic muscle dysfunction — including myofascial pain, fibromyalgia, and post-exercise muscle damage — involves persistent disruptions to sarcomere geometry (desmin disruption, Z-disc streaming), ion channel expression changes that alter baseline EMF geometry, and Ca^{2+} handling abnormalities that shift the contraction window baseline [33]. Treating these conditions purely through energy metabolism (nutrition, rest) is incomplete. Interventions that target ionic and structural geometry — specific electrolyte protocols, NMES, controlled eccentric rehabilitation to remodel sarcomere length distributions, and pharmacological restoration of channel geometry — address the actual site of dysfunction. The geometry was the problem. The geometry is the fix.

Discussion

The field-geometry framework proposed here does not discard biochemistry. It reframes where chemistry sits in the causal hierarchy. Chemistry sets the energy budget. Geometry determines how that budget is spent. Every biochemical event in the cross-bridge cycle — Ca^{2+} binding to troponin C, Pi release from myosin, ATP hydrolysis, ADP dissociation — has a geometric consequence. The framework argues that those geometric consequences are the primary determinants of force production, directionality, efficiency, and failure, and that biochemical accounts that do not explicitly represent the geometry are mechanistically incomplete.

This is not a merely verbal reframing. The framework generates distinct, testable predictions. First, contraction efficiency — force per ATP consumed — should correlate with lattice geometry precision, not just ATP availability. Conditions that disturb lattice geometry (altered osmotic pressure, pathological lattice spacing, sarcomere length non-uniformity) should reduce efficiency even when ATP is abundant, and this reduction should be quantitatively predictable from geometric parameters. X-ray diffraction studies of contracting muscle by Reconditi and colleagues have already demonstrated that lattice spacing changes alter cross-bridge force production in geometry-consistent ways [18], providing a methodological template for testing this prediction more broadly.

Second, the framework predicts that fatigue interventions targeting ionic geometry should outperform those targeting energy metabolism alone when ionic geometry drift is the dominant failure mechanism. This prediction is testable: fatigue protocols designed to produce K^+ -accumulation-dominant failure (high-frequency stimulation in isolated T-tubule preparations) should respond better to interventions that restore K^+ gradients than to interventions that replenish ATP precursors. The existing literature on T-tubule K^+ accumulation and fatigue [22, 30] provides a starting point, but direct geometry-restoration experiments are needed.

Third, channelopathies — pathologies characterized by abnormal ion channel geometry or function — should produce predictable, geometry-consistent contraction deficits. Periodic paralysis (both hypokalemic and hyperkalemic forms) involves mutations in Na^+ or Ca^{2+} channels that alter the EMF geometry of the sarcolemma in specific, mappable ways [34]. The framework predicts not just that these mutations impair contraction, but that they impair it in a specific geometric pattern: the nature of the EMF disruption should predict the pattern of contraction window failure, the fiber types affected, and the therapeutic response to ionic correction.

The framework extends naturally to other muscle types. Cardiac muscle uses the same actin-myosin geometry, the same DHPR voltage sensor, and the same troponin switch — but Ca^{2+} -induced Ca^{2+} release (CICR) dominates over direct DHPR-RyR2 mechanical

coupling, and the propagation geometry is driven by gap junctions rather than sarcolemmal action potential conduction [13]. The geometry-first account applies directly: cardiac excitation is a geometric wave propagating through a syncytium, injecting Ca^{2+} -mediated tension across the ventricular wall. Cardiac failure — including ischemia-driven Ca^{2+} overload — is geometry drift with specific ionic and structural signatures that the framework characterizes precisely.

Smooth muscle presents a slower version of the same framework. Contraction windows are longer (latch-bridge formation allows sustained force at low ATP cost), geometry reset is slower (smooth muscle myosin has lower ATPase rate), and regulatory geometry is different (calmodulin-MLCK rather than troponin-tropomyosin controls thin filament access) [35]. But the geometry-first account holds: smooth muscle contraction is triggered by Ca^{2+} -calmodulin geometry change, sustained by latch-bridge geometry, and regulated by myosin phosphorylation state — all geometric descriptions.

The framework requires experimental quantification in several areas. The precise tolerance bands for cross-bridge geometry matching under varying ionic conditions have not been comprehensively mapped. The relationship between T-tubule K^+ geometry drift and action potential propagation failure needs more precise quantification at the single T-tubule level. The contribution of sarcomere length non-uniformity to whole-muscle fatigue — relative to ionic drift — requires simultaneous measurement of sarcomere length distributions and force in fatiguing fibers. These are tractable experiments, and the framework provides clear predictions that each of them can test.

Conclusion

Muscle contraction is a geometric event powered by chemistry. The ion field creates the tension. The filament lattice defines the geometry. Productive cross-bridge formation requires alignment within geometric tolerance. Propagation spreads the geometric

activation wave across the fiber volume. ATP resets the geometry for the next cycle. Failure is drift. Restoration is correction.

This framework is not a metaphor. It is a mechanistically precise description of events that already have spatial coordinates, angular measurements, and displacement vectors attached to them. The nanometer-scale geometry of actin's helix pitch, myosin's lever arm rotation, tropomyosin's azimuthal displacement, and the T-tubule's waveguide architecture are measured facts. The voltage-sensor's gating charge movement is a measured physical displacement. The sarcomere's length-tension curve is a measured geometric consequence. The geometry was always there. The field-geometry framework simply names it, traces it through the full contraction cycle, and uses it to explain contraction, fatigue, spasm, and restoration under a single coherent account.

If contraction is geometry, then understanding muscle means measuring geometry — not just chemistry. That shift in emphasis is the practical consequence of this framework, and it points directly toward new experimental approaches, new therapeutic targets, and a cleaner causal story for one of biology's most fundamental processes.

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